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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXP.NO: 4</b> | <b>Create a web form that accepts a user's name and age.<br/>Write a Servlet to process the form data and display it<br/>back on the browser.</b> |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

### AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

### ALGORITHM:

#### 1. Input:

- The user fills out a web form with their **name** and **age** and submits it.

#### 2. Form Handling and Servlet Processing:

##### 1. Create the Web Form:

- Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
- Add a **submit button** to the form.
- Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).

##### 2. Create the Servlet Class:

- Write a Servlet that processes the **form data** submitted by the user.
- The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.
- The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").

##### 3. Process the Data in the Servlet:

- Extract the name and age values submitted from the form.
- **Validate the input** (e.g., check if the age is a valid number).
- Prepare a **response** that will display the user's **name** and **age** back on the browser.

#### 4. Send Response to the Browser:

- The Servlet will send a **response** to the browser containing a message like:
  - "Hello, [name]! You are [age] years old."
- The response will be displayed to the user in the browser.

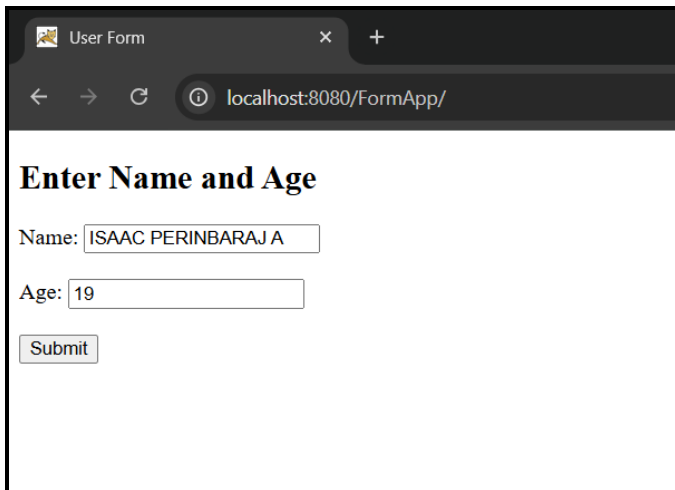
#### 5. Deploy the Servlet:

- Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
- Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.

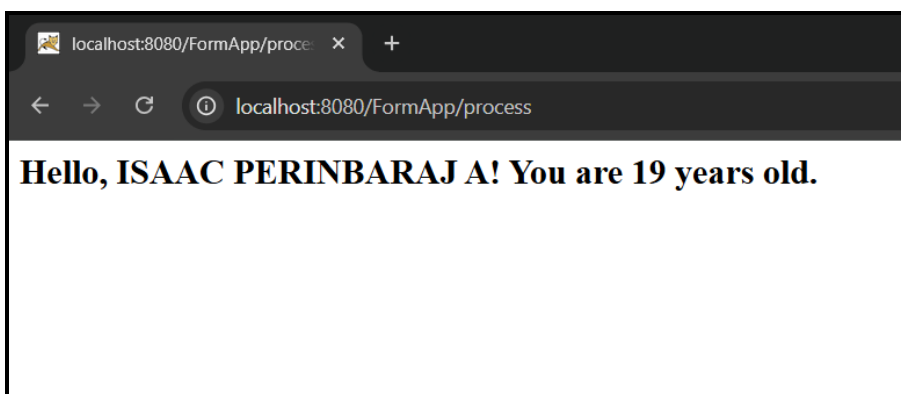
#### 6. Test the Application:

- Open the web browser, fill in the form with the name and age, and submit it.
- The Servlet will process the data and display the message back to the user in the browser.

### Output:



A screenshot of a web browser window. The title bar says 'User Form'. The address bar shows 'localhost:8080/FormApp/'. The page content has a heading 'Enter Name and Age'. Below the heading, there are two input fields: 'Name: ISAAC PERINBARAJ A' and 'Age: 19'. At the bottom of the form is a 'Submit' button.



A screenshot of a web browser window. The title bar says 'localhost:8080/FormApp/process'. The address bar shows 'localhost:8080/FormApp/process'. The page content displays the message 'Hello, ISAAC PERINBARAJ A! You are 19 years old.'

## HTML File (index.html):

```
<!DOCTYPE html>
<html>
<head>
    <title>User Form</title>
</head>
<body>
    <h2>Enter Name and Age</h2>
    <form action="process" method="post">
        Name: <input type="text" name="name"><br><br>
        Age: <input type="text" name="age"><br><br>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

## Java File (MyServlet.java):

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class MyServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

        String name = request.getParameter("name");
        String age = request.getParameter("age");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html><body>");

        try {
            int a = Integer.parseInt(age);
            out.println("<h2>Hello, " + name + "! You are " + a + " years
old.</h2>");
        } catch (Exception e) {
            out.println("<h2>Invalid age!</h2>");
        }

        out.println("</body></html>");
    }
}
```

### File (web.xml) :

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
    version="5.0">

    <servlet>
        <servlet-name>MyServlet</servlet-name>
        <servlet-class>MyServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>MyServlet</servlet-name>
        <url-pattern>/process</url-pattern>
    </servlet-mapping>

</web-app>
```

### RESULT:

A Servlet is created to process the HTML form data and display it back on the browser.